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Math 3108 – Linear Algebra
Exam 3 – Part A
Prof. Runnion
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Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned off or completely silenced (i.e. not on vibrate) for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 4) Failure to follow directions specific to a problem will result in the loss of points.
- 5) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 6) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 7) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 8) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 9) If you need extra room, you may use the back of the previous page. However, you must indicate you are doing so by clearly writing "BPP" on the relevant problem.
- 10) This packet has four sheets of paper, including this cover page. Part B has three sheets of paper, including a cover page. Do NOT remove the staple or remove any sheet from this packet.
- 11) Once this exam begins, you will have 50 minutes to complete your solutions.

Page	1	2	3	Part A	Part B	Sum
Points Earned	15	18	18	51	35	86
Maximum Points	21	24	18	63	37	100

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

Name: _____

Multiple Choice and True/False. Circle the correct answer. 3 points each.

1. Let A be a 5×5 matrix and let λ be an eigenvalue of A with multiplicity 3. The *minimum* dimension of the eigenspace of A corresponding to λ is _____.
0 ☒ 1 2 3 4 5
2. Let A be a 5×5 matrix and let λ be an eigenvalue of A with multiplicity 3. The *maximum* dimension of the eigenspace of A corresponding to λ is _____.
0 1 2 ☒ 3 4 5
3. Every linearly independent set is orthogonal.
a) True ☒ b) False
4. If A is a 5×3 matrix, what is the smallest possible dimension of $\text{Nul } A$?
☒ 0 1 2 3 4 5
5. If the null space of a 5×6 matrix is 4-dimensional, what is the dimension of $\text{Row } A$?
0 1 ☒ 2 3 4 5
6. If a set of 5 vectors spans a 5-dimensional subspace W of $\mathbb{R}^{21}$, then the set is _____ a basis for W .
☒ a) Always b) Sometimes ☒ c) Never
7. Let $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{y}$ be vectors in $\mathbb{R}^3$, and let k be a nonzero scalar. Let $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$. If $\text{proj}_W \mathbf{y} = k\mathbf{v}_2$, then which of the following must be true?
☒ a) $\mathbf{y}$ is orthogonal to $\mathbf{v}_1$ ☒ b) $\mathbf{y}$ is orthogonal to $\mathbf{v}_2$
c) $\mathbf{y}$ is a multiple of $\mathbf{v}_1$ d) $\mathbf{y}$ is a multiple of $\mathbf{v}_2$

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Open Answer. Show ALL your work!

8. Let $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$, where $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$, and let $\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$.

$$\frac{1+9-10}{1+9+4} = 0 \bar{u}_1$$

$$\frac{5+3+20}{25+1+16} = \frac{28}{42} = \frac{14}{21} = \frac{2}{3} \bar{u}_2$$

- a) (8 points) Write $\mathbf{y}$ as the sum of a vector $\hat{\mathbf{y}}$ in W and a vector $\mathbf{z}$ orthogonal to W .

$$\hat{\mathbf{y}} = \text{proj}_W \bar{\mathbf{y}} = \frac{\bar{\mathbf{y}} \cdot \bar{\mathbf{u}}_1}{\bar{\mathbf{u}}_1 \cdot \bar{\mathbf{u}}_1} \bar{\mathbf{u}}_1 + \frac{\bar{\mathbf{y}} \cdot \bar{\mathbf{u}}_2}{\bar{\mathbf{u}}_2 \cdot \bar{\mathbf{u}}_2} \bar{\mathbf{u}}_2 = \frac{1(1)+3(3)+5(-2)}{1^2+3^2+(-2)^2} \bar{\mathbf{u}}_1 + \frac{1(5)+3(1)+5(4)}{5^2+1^2+4^2} \bar{\mathbf{u}}_2$$

$$\bar{\mathbf{z}} = \bar{\mathbf{y}} - \hat{\mathbf{y}} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} - \begin{bmatrix} 10/3 \\ 2/3 \\ 8/3 \end{bmatrix} = \begin{bmatrix} -7/3 \\ 7/3 \\ 7/3 \end{bmatrix} = 0 \bar{\mathbf{u}}_1 + \frac{2}{3} \bar{\mathbf{u}}_2 = \frac{2}{3} \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 10/3 \\ 2/3 \\ 8/3 \end{bmatrix} = \hat{\mathbf{y}}$$

$$\begin{aligned} 3/3 - 10/3 \\ 9/3 - 2/3 \\ 15/3 - 8/3 \end{aligned}$$

8

$$\bar{\mathbf{y}} = \hat{\mathbf{y}} + \bar{\mathbf{z}} = \begin{bmatrix} 10/3 \\ 2/3 \\ 8/3 \end{bmatrix} + \begin{bmatrix} -7/3 \\ 7/3 \\ 7/3 \end{bmatrix}$$

- b) (4 points) Find the distance from $\mathbf{y}$ to the plane in $\mathbb{R}^3$ spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$.

$$\|\bar{\mathbf{z}}\| = \sqrt{(-7/3)^2 + (7/3)^2 + (7/3)^2} = \sqrt{49/9 + 49/9 + 49/9}$$

$$\boxed{\frac{7}{3}\sqrt{3}}$$

4

9. (12 points) Find the eigenvalues and associated eigenvectors of $A = \begin{bmatrix} 0 & 1 \\ -8 & 4 \end{bmatrix}$

$$\det(A - \lambda I) = 0 : \det \left(\begin{bmatrix} 0-\lambda & 1 \\ -8 & 4-\lambda \end{bmatrix} \right) = [(-\lambda)(4-\lambda) + 8] = 4\lambda + \lambda^2 + 8$$

$$\lambda^2 + 4\lambda + 8$$

$$-4 \pm \sqrt{4^2 - 4(1)(8)}$$

$$2(1)$$

$$\frac{-4 \pm \sqrt{16 - 32}}{2}$$

$$2$$

$$\frac{-4 \pm \sqrt{-16}}{2} = \frac{-4 \pm 4i}{2}$$

$$\boxed{\lambda = -2 \pm 2i}$$

$$(A - \lambda I)\bar{\mathbf{x}} = 0 : \text{For } -2 + 2i = \lambda$$

$$4 - (-2 + 2i) = 6 - 2i$$

$$0 - (-2 + 2i) = 2 - 2i$$

$$\begin{bmatrix} -2-2i & 1 \\ -8 & 6-2i \end{bmatrix} \xrightarrow{\substack{R_1 \leftrightarrow R_2 \\ R_2 \cdot \frac{1}{2}}} \begin{bmatrix} 4 & -3+2i & 0 \\ -2-2i & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -3+2i & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} x_1 &= \frac{3}{4} - \frac{1}{2}i \\ x_2 &\text{ Free} \end{aligned}$$

$$\text{Let } x_2 = 1$$

$$\bar{\mathbf{x}} = \begin{bmatrix} 3/4 - 1/2i \\ 1 \end{bmatrix}$$

$$\text{For } -2+2i$$

And

$$\bar{\mathbf{x}} = \begin{bmatrix} 3/4 + 2i \\ 1 \end{bmatrix}$$

$$\text{For } -2-2i$$

-4

-2

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10. (18 points) Diagonalize $A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}$, if possible. If it is not possible, clearly explain

why. Note that the eigenvalues of A are $\lambda = 1$ and $\lambda = 5$.

$$(A - \lambda I)\bar{x} = 0 : \lambda = 1 \quad \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 1 & 2 & -1 & 0 \\ -1 & -2 & 1 & 0 \end{array} \right] \begin{array}{l} \rightarrow -R_1 + R_2 \rightarrow R_2 \\ \rightarrow R_1 + R_3 \rightarrow R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\bar{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$x_1 = -2x_2 + x_3$$

x_2 Free

x_3 Free

$$\lambda = 5 : \left[\begin{array}{ccc|c} -3 & 2 & -1 & 0 \\ 1 & -2 & -1 & 0 \\ -1 & -2 & -3 & 0 \end{array} \right] \begin{array}{l} \rightarrow \text{Swap } R_1 \text{ and } R_2 \\ \rightarrow 3R_1 + R_2 \rightarrow R_2 \\ \rightarrow R_1 + R_3 \rightarrow R_3 \end{array} \left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ -3 & 2 & -1 & 0 \\ -1 & -2 & -3 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & -4 & -4 & 0 \end{array} \right] \begin{array}{l} \rightarrow \frac{1}{4}R_2 \\ \rightarrow \frac{1}{4}R_3 \end{array} \left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \rightarrow -R_2 + R_3 \rightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow 2R_2 + R_1 \rightarrow R_1 \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} x_1 = -x_3 \\ x_2 = -x_3 \\ x_3 \text{ Free} \end{array}$$

$$\bar{x} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \quad \text{with} \quad P = \begin{bmatrix} -2 & 1 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$